Junghoon Chung

Seattle | junghoon@uw.edu | 540-808-5715 | |

Education

University Of Washington, Ph.D in Industrial and Systems Engineering — Sep 2025 -

- Advisor: Dr. Ji-Eun Kim (HAS LAB)

Virginia Tech, M.S in Industrial and Systems Engineering — Aug 2023 - May 2025

- Advisor: Dr. Sol Lim (OEB LAB)
- Certificate: Human-Computer Interaction
- Thesis: Understanding and Predicting Sit-Stand Desk Usage Patterns and Willingness among Knowledge Workers: A Data-Driven Approach (Link)

Yonsei University, B.S in Industrial Engineering — Mar 2017 - Feb 2023

Research Experience

Research Assistant, HAS Lab, University of Washington — Sep 2025 - Present
Research Assistant, OEB Lab, Virginia Tech — Aug 2023 - May 2025

Research Interest

- Human Pattern Prediction using Context and Complex Data
- Data-driven Human Performance Modeling using ML/DL Algorithms
- Human Performance Enhancement using Personalized Digital Intervention

Publications

JOURNAL

[In Progress] Chung, J*., Prendez, D*., Higgins, EA. & Kim, JE., Assessing Clinician Vigilance using Physiological Measures and Dynamic Bayesian Network Modeling **:Co-first author*

[In Progress] Chung, J., Hu, D., Lee, SW. & Lim, S., Predictive Model of Sit-Stand Desk Usage Patterns among knowledge workers and Customization based on Behavior Clustering.

[P1] Hu, D., **Chung, J.,** Vargas-Diaz, D., Lee, SW. & Lim, S., Toward Context-Aware and Personalized Sit-Stand Desk Interventions: Insights from a Field Observational Study.
International Journal of Human-Computer Studies

CONFERENCE

[C2] Chung, J. & Kim, JE., Real-time Vigilance Monitoring via Biosignals: A Scoping Review.
Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2026

[C1] Chung, J., Hu, D., Vargas-Diaz, D., Lee, SW., & Lim, S., Understanding the Context of Sit-Stand Desk Usage for Promoting Healthy and Productive Behaviors.
Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2025

WORKSHOP

[W1] Hu, D., **Chung, J.,** Vargas-Diaz, D., Lim, S., & Lee, SW. Developing Context-Aware Sit-Stand Desks for Promoting Healthy and Productive Behaviors. CHI 2024 Office Wellbeing by Design Workshop.

Projects

University of Washington
Project 1 : Assessing Vigilance with Biometric Data for Healthcare Providers (P.I: Ji-Eun Kim)

- Processed physiological data from participants and constructed ML models to assess vigilance.
- Integrated multimodal biometric signals including heart rate variability, skin conductance, temperature, and eye tracking features to predict reaction time and performance degradation.

Project 2 : Driver Response to Connected Vehicle Alerts at Pedestrian Crossings – City of Bellevue (P.I: Ji-Eun Kim)

- Developed study protocol, IRB documentation, and questionnaires in collaboration with the City of Bellevue, T-Mobile, and Applied Information.
- Collected multimodal data including vehicle kinematics, Empatica E4 physiological signals, in-vehicle video, and contextual interview responses during on-road driving sessions with human participants.
- Processed and analyzed physiological (EDA, heart rate) and driving behavior (speed, deceleration, yielding) data to evaluate the effectiveness of smartphone-based pedestrian alerts.
- Reconstructed interpretable behavioral and physiological features by integrating observer-logged event timestamps with synchronized sensor data streams, and developed predictive models to assess driver alertness and yielding behavior at pedestrian crossings.

Project 3: Multi-Stage Behavioral Modeling of Driver Takeover During Automated Driving (P.I: Ji-Eun Kim, Linda Ng Boyle)

- Developing a multi-stage behavioral framework to characterize driver takeover as a sequential process consisting of Attention Reorientation, Control Acquisition, and Control Stabilization
- Processed synchronized eye-tracking, driving simulator, and contextual data to reconstruct interpretable temporal behavioral features representing each stage of the takeover process.
- Identified and operationalized representative behavioral indicators through theory-driven feature engineering, redundancy analysis, and empirical evaluation to improve interpretability and reduce model complexity.
- Evaluated the relationships between behavioral stages and developed predictive models to assess how attention reorientation and early control behaviors contribute to successful takeover performance.

Project 4: Real-Time Vigilance Monitoring Interface (P.I: Ji-Eun Kim)

- Developing a real-time vigilance-monitoring system that streams EmotiBit physiological data, starting with heart rate, electrodermal activity, and photoplethysmography, into trained models for continuously updated vigilance estimates.
- Designing live-learning workflows to personalize vigilance predictions as individual data accumulate over time.
- Identified and operationalized representative behavioral indicators through theory-driven feature engineering, redundancy analysis, and empirical evaluation to improve interpretability and reduce model complexity.
- Conducting user interviews to design monitoring feedback and planning lab and prospective classroom field studies of performance, usability, and perceptions of monitoring.

Virginia Tech

Project 1 : Context-Aware Sit-Stand Intervention for Promoting Healthy Behaviors in Knowledge Workers (P.I: Sangwon Lee, Sol Lim)

- Developed an end-to-end data science pipeline to predict worker context patterns by integrating multi-source data, implementing a zero-shot text classification model, performing mixed-method hypothesis testing, and building time-series forecasting models.

Project 2 : Enabling Multifaceted Collaboration between Humans, Robots, and Remote Workers (P.I: Sol Lim)

- Executed a comprehensive user study by acquiring multi-modal objective data (EMG, EEG, IMU, eye-tracking) and subjective questionnaires to generate and test hypotheses with time-series analysis of posture logs.

Project 3 : Responsible Artificial Intelligence (AI)-Based Biomechanical Exposure Assessment (P.I: Sol Lim)

- Designed and executed experiments to collect and analyze 3D motion capture and physiological status to build an dataset for de-biased AI-based biomechanical exposure assessments.

Course Projects

- Human performance on perceiving facial expression with random distributed face
- Human performance and mental demand comparison with subjective scores and K-means clustering methods
- Verification of new gamifying feature for Pl@ntNet through full usability engineering cycle
- Seoul daycare center community formation using Clustering methods

- Review reliability analysis by BERT sentiment analysis and Clustering
- Quality management of delivery system in Korea using QFD and Conjoint Analysis
- Semi-supervised learning for rock-paper-scissor-spock-lizard image classification

Honors and Awards

University of Washington
- Dean's Fellowship, College of Engineering, 2025 - 2029
- Top Reviewer, HFES ASPIRE 2025

Yonsei University
- Merit-based Scholarship (Academic Excellence)
- Social-Innovation Capability Video Contest 2nd Place – Data mining

Extra Activities

Invited Talks
- From Vigilance Prediction to Adaptive Interventions: Data-Driven Approaches to Managing Fatigue in Safety-Critical Work, PSHFES Symposium 2026

University of Washington
- Member of HFES Student Chapter at University of Washington

Virginia Tech
- Member of HFES Student Chapter at Virginia Tech
- Member of Center for Human-Computer Interaction at Virginia Tech
- Member of KSEA, Southern Virginia Chapter (Korean-American Scientists and Engineer Association)

Yonsei University
- President of Industrial Engineering Student Council
- Member of YBIGTA, Student Big Data Society

Other
- Republic of Korea ARMY, 22nd division, Coast Guard & Administrative Specialist

Links

LinkedIn
- www.linkedin.com/in/junghoon-chung

Google Scholar
- https://scholar.google.com/citations?user=_yF8Yt0AAAAJ&hl=en&oi=sra

Personal Website
- https://junghoon-chung.github.io/